Factory Relay: On-Device Learning for Multilingual Factory Communication

Proposed extension of the animated Factory Relay game in PurePolyglot. Automated local translation, cross-device play, personal model training and a guarded laboratory demonstration are planned work.

The industrial problem

Factory operators and maintenance coordinators must agree on the equipment, action, quantity, timing and release conditions in a message. Multilingual speech, regional expressions, code-switching and workplace shorthand can complicate this agreement in global and geographically-spread organizations. An apparently fluent translation may still change a negation or identify the wrong machine. The target workflow is fault report → clarification → confirmed maintenance handover. Health and Safety Executive (HSE) identifies information exchange and cross-checking as important in industrial handovers [1].

Company evidence and Smart Factory relevance

DSV is a global 3PL provider operating "Smart Warehouses" and "Fulfilment Factories" in South Korea (Seoul, Incheon, Busan) using Industry 4.0 automation, robotics, and AI tech found in smart manufacturing.

A DSV (a global transport and logistics company) manager reportedly found some translated Mandarin chats difficult to understand during warehouse support. Supplied screenshots show mixed-language messages and label-printing coordination. This motivates inquiry; it does not prove a manufacturing incident, dialect-specific error or measured loss. The first-month review will seek a company-reviewed workflow and anonymized examples. If the case remains warehousing, confirm its smart-logistics eligibility with the supervisor.

The proposed game connects digital equipment states, work orders and human decisions in a simulated conveyor, processing and labeling cell. Manufacturing simulation has an established research and training basis [9]. The contribution to test is whether locally adapted language support preserves operational meaning during this process.

Research question and scope

Can speaker-reviewed local memory and optional personal adapters improve multilingual operational understanding over the same frozen model, while keeping personal recordings and training examples on the user's device?

Players select source and recipient languages or dialects; original wording is retained. Translation may be into an underrepresented variety, as well as from it. Accepting an input is not proof of accurate coverage: evaluate one community-reviewed language pair or variety first. The system should ask for clarification when meaning is uncertain.

Evidence basis and current status

Automated speech recognition (ASR) racial disparities are documented by Koenecke et al. (2020), but findings for particular systems and populations cannot establish universal error rates. Streaming translation already exists [7]. This project tests local adaptation and operational meaning rather than claiming to invent live translation. The current game supplies animation, scripted scenarios, and local text play; the capabilities below require implementation and evaluation.

Local architecture and model delivery

Use the existing browser interface with an installed local companion for inference, encrypted storage and training. On first launch, download a versioned compatible model pack; later launches verify and reuse the cache. Qwen3-1.7B is a candidate text model [3]; Qwen3-ASR-0.6B-hf is a candidate voice extension [4]. These are starting points to benchmark, not validated dialect translators. We still have to check license, language support, memory and latency on the actual device.

The pack distinguishes a quantized inference artifact from the checkpoint needed for training. We will keep the base model frozen and train a small personal LoRA adapter using PEFT [5]. We still start with a capable laptop; phone inference and retrieval can follow, with phone training enabled only after validation. Browser inference is possible [11]; a native companion is the practical initial route for this training workload.

Offline operation and two-device play

After installation and model download, gameplay, local translation, review and eligible adapter training shall work without internet. Precache the app and scenarios [15]; install compatible model/tokenizer files and runtimes on each participating device, and enable local-only loading [16]. Missing resources trigger setup guidance, never silent cloud fallback.

Two physical devices shall play over the same Wi-Fi router or hotspot with internet disabled. For the first pilot, a laptop hosts the local room and signaling service; the second device joins through a locally trusted HTTPS address and room code/QR. Use encrypted peer channels or a local relay if needed [6], without public signaling/STUN/TURN dependencies. Each device retains its own model and training data. The local network must permit peer access; localhost addresses cannot identify another device. Single-device play needs no network.

Current status (8 October 2026): pass-and-play and same-browser paired tabs exist; two-device play remains planned. One already-loaded round passed with networking disabled. Offline restart, local AI/ training and LAN multiplayer require further validation; the development link needs its local server running.

Translation and adaptation loop

During play: preserve typed text or locally recorded audio; retain the ASR draft separately; retrieve approved terminology and user-selected language-profile examples; produce a translation in the recipient’s chosen variety; display the original and uncertainties. Profiles are selected and reviewed by speakers, not inferred demographic stereotypes. No silent cloud fallback or unapproved upload is permitted.

After each session: choose keep/delete → review corrections → select examples → update memory and/or train → evaluate candidate → activate or roll back. Memory updates change retrieval, not model weights. Actual training uses reviewed labels: audio → corrected transcript for ASR; text → intended meaning/target rendering for interpretation. Target-language labels require a competent bilingual or target-community reviewer. Do not learn blindly from model outputs or hidden game answers.

Offer the choice after every game, but queue examples if data or hardware is insufficient. Hold out review examples, test critical meanings and previously learned skills, and activate an adapter only between sessions. The user can compare, export or delete it. Few verified examples may help retrieval more than training; improvement is an empirical question.

Cross-device play and data ownership

Use encrypted WebRTC data channels with signaling and STUN/TURN support [6]. Exchange selected messages and game events; keep raw recordings and personal training corpora local. Recipients can retain messages they receive. Protect the local vault with OS-backed key storage and audit existing app telemetry. Sharing adapters or federated updates is optional future work; blockchain is unnecessary for the core study.

Physical demonstration milestone

If suitable hardware is available, connect a supervised laboratory conveyor or PLC through a restricted gateway. Translation proposes an action; a human confirms it; deterministic permissions, fresh-state checks and interlocks authorize execution. The controller handles network loss and independent stopping. OPC UA may provide the interface [10]; it does not by itself make model-generated control safe. No deployed industrial-machine control is claimed.

Evaluation and technical boundaries

Compare the same frozen model, model + local memory, and model + memory + personal adapter using matched, held-out scenarios and counterbalanced order. Score correct operational decisions, critical-field errors, unsupported additions, clarification time, latency and device resource use. Keep hidden scoring state out of model context. Report attempted wrong actions separately from simulator or gateway blocks. Offline acceptance tests: after provisioning, block internet access and cold-start the app; complete and export a round; run local translation, review, eligible adapter training and checkpoint reload/rollback; then restart and complete a round on two physical devices with internet disabled, testing a router and hotspot separately. Verify shared state, message order, disconnect/rejoin and rejection of duplicate actions. Audit external requests and local data retention. Report these checks separately from translation accuracy.

For speech, also report WER/CER against dialect-faithful references. Treat SAER as an explicitly defined supporting score: the supplied CA-IEDI manuscript uses semantic cosine distance; SwitchLingua combines semantic distance with WER/CER [8]. These definitions are not interchangeable. Human-reviewed operational fields remain primary. Acoustic profiling and raw-versus-denoised audio comparisons are exploratory; no accuracy gain is assumed.

Deliverables and semester sequence

Weeks 1–4: company problem sheet, reviewed language scope, device benchmark and updated demonstration. Weeks 5–8: local model-pack delivery, translation, laptop-hosted two-device rooms, local pairing and consent-based capture. Weeks 9–12: correction interface, memory updates, genuine optional adapter training and rollback. Weeks 13–15: controlled evaluation, hardware-dependent guarded lab demonstration, report and presentation. Two teaching leads review industrial realism and language/AI evaluation. Offline delivery adds a readiness check for cached assets and model packs, a two-device setup guide, and a demonstration/report with internet disabled within the same semester phases.

Principal risks and responses

Limited device capacity → smaller model or queue training. Poor target-variety coverage → narrow the pilot and recruit reviewers. Overfitting or harmful normalization → held-out checks and rollback. Private-data exposure → local storage, network audit and explicit sharing. Hardware or company access delays → report the unmet milestone and retain the reproducible simulator evaluation. Neither a patent nor a low-WER threshold is promised. Offline risks: cache eviction or missing files require verified local packs and readiness checks; LAN isolation or hotspot/firewall restrictions require connection checks and a pass-and-play fallback; host loss pauses shared play; interrupted training retains the last validated adapter.

Selected references — linked citations

[1] HSE. Shift handover: information exchange and cross-checking.

[2] Koenecke et al. (2020). Racial disparities in automated speech recognition. PNAS.

[3] Qwen. Qwen3-1.7B: official model card and license.

[4] Qwen. Qwen3-ASR-0.6B-hf: official speech model card.

[5] Hugging Face. PEFT: quantization and adapter training.

[6] WebRTC. Peer connections, signaling and connectivity.

[7] Seamless Communication et al. Seamless: multilingual, expressive and streaming speech translation.

[8] SwitchLingua (2025). Dataset and semantic-aware evaluation. NeurIPS Datasets and Benchmarks.

[9] McLean et al. (2005). Manufacturing simulation and gaming. NISTIR 7256.

[10] OPC Foundation. OPC UA security model: Part 2, section 4.

[11] MLC. WebLLM documentation: browser inference.

[12] J. Fiset, T. Nguyen, and M. Roberts, “The effects of language-related misunderstanding at work,” Journal of Management, vol. 49, no. 2, pp.345–372, 2023.

Literature review and evaluation of previous work

Selected prior work and implications for this proposal. Studies use different tasks and datasets; their scores are not directly comparable. The final column states proposed evaluation needs, not findings from those papers.

Previous work	Contribution	Evaluation / evidence	Implication for Factory Relay
Koenecke et al. PNAS, 2020 [2]	Audits disparities across five commercial speech-recognition services.	US interview recordings: mean WER 0.35 for Black speakers versus 0.19 for white speakers. Historical, corpus-specific results.	Evaluate the selected speaker varieties separately; test whether transcription errors change operational meaning.
Seamless Communication 2023 [7]	Streaming multilingual translation; related components preserve vocal style and prosody.	Automatic latency, robustness and prosody evaluations; human assessments of meaning preservation, naturalness and expressivity.	Live translation is established. Test local, reviewed adaptation and factory decisions for the selected varieties.
Lewis et al. RAG, 2020 [13]	Combines a language generator with retrieved external knowledge.	Open-domain question answering and generation benchmarks; reports stronger factual generation than a parametric-only baseline.	Motivates local phrase retrieval. Its results do not establish accurate dialect mediation; test approved examples against no-memory inference.
Hu et al. LoRA, 2021 [14]	Trains low-rank weight updates while freezing the pretrained base model.	Language-task quality, trainable parameters, memory and throughput; matches or exceeds full fine-tuning in reported experiments.	Supports personal adapters. Measure held-out meaning, forgetting and training cost; one-session improvement on edge devices remains unproven.
Xie et al. SwitchLingua, 2025 [8]	Code-switching text/audio dataset and semantic-aware ASR evaluation.	Benchmarks ASR using WER/CER, semantic similarity and SAER; SAER combines semantic distance with lexical error.	Useful evaluation reference, not proof of factory translation quality. Retain bilingual review and checks of action, negation and quantity.
McLean et al. NISTIR 7256, 2005 [9]	Architecture connecting manufacturing simulations and games for research, testing and training.	Presents architecture and use cases, rather than a controlled dialect-translation or communication-improvement trial.	Supports the game setting. Validate scenario behavior and measure communication-supported operational decisions.

Factory Relay will compare the same frozen model, model + reviewed local memory, and model + memory + personal adapter. Use held-out sessions to measure operational meaning, translation adequacy, latency, resource use and retention. This selected review supports the study design; it does not establish patent novelty.

Additional references for the table and offline operation

[13] Lewis et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. NeurIPS.

[14] Hu et al. (2021 preprint). LoRA: Low-Rank Adaptation of Large Language Models.

[15] Chrome for Developers. Workbox precaching: storing and serving application assets offline.

[16] Hugging Face. Transformers installation: offline mode and local model loading.